THE EXISTENCE PART OF THE VALUATIVE CRITERION MAY BE TESTED WITH FINITE SEPARABLE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author states that only the implication (1) implies (2) has to be shown, and the proof carries that implication out in full; the converse direction is the formal restriction of the same condition.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The shape of the diagram in the valuative criterion. Definition definition-valuative-criterion, used at proof lines 8451-8458. Fixes the commutative square that has to be completed.
- Generalizations lift along flat morphisms of schemes. Morphisms, Lemma morphisms-lemma-generalizations-lift-flat, used at proof lines 8465-8469. Produces the generalization p' of p mapping to the generic point of Spec(A').
- Existence of a valuation ring dominating a given local subring. Algebra, Lemma algebra-lemma-dominate, used at proof lines 8501-8504. Supplies the valuation ring A_1 inside kappa(q') that dominates B_q.
- Base change of surjective etale morphisms of algebraic spaces. used at proof lines 8459-8463. Makes Spec(A') x_X U -> Spec(A') surjective etale, so that the point p exists.

Source reference. Original lemma 00IA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 672–672. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4581–4581. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent*

(1) *f satisfies the existence part of the valuative criterion as in Definition 03IX,*
(2) *f satisfies the existence part of the valuative criterion as in Definition 03IX modified by requiring the extension K'/K to be finite separable.*

Proof. We have to show that (1) implies (2). Suppose given a diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

as in Definition 03IX with $K \subset K'$ arbitrary. Choose a scheme U and a surjective étale morphism $U \to X$. Then

$$\mathrm{Spec}(A') \times_X U \longrightarrow \mathrm{Spec}(A')$$

is surjective étale. Let p be a point of $\mathrm{Spec}(A') \times_X U$ mapping to the closed point of $\mathrm{Spec}(A')$. Let $p' \rightsquigarrow p$ be a generalization of p mapping to the generic point of $\mathrm{Spec}(A')$. Such a generalization exists because generalizations lift along flat morphisms of schemes, see Morphisms, Lemma 03HV. Then p' corresponds to a point of the scheme $\mathrm{Spec}(K') \times_X U$. Note that

$$\mathrm{Spec}(K') \times_X U = \mathrm{Spec}(K') \times_{\mathrm{Spec}(K)} (\mathrm{Spec}(K) \times_X U)$$

Hence p' maps to a point $q' \in \mathrm{Spec}(K) \times_X U$ whose residue field is a finite separable extension of K. Finally, $p' \rightsquigarrow p$ maps to a specialization $u' \rightsquigarrow u$ on the scheme U. With all this notation we get the following diagram of rings

$$\begin{array}{ccccc} \kappa(p') & \longleftarrow & \kappa(q') & \longleftarrow & \kappa(u') \\ & \nwarrow & \uparrow & & \uparrow \\ & & \mathcal{O}_{\mathrm{Spec}(A') \times_X U, p} & \longleftarrow & \mathcal{O}_{U,u} \\ & & \uparrow & & \\ K' & \longleftarrow & A' & \longleftarrow & A \end{array}$$

This means that the ring $B \subset \kappa(q')$ generated by the images of A and $\mathcal{O}_{U,u}$ maps to a subring of $\kappa(p')$ contained in the image B' of $\mathcal{O}_{\mathrm{Spec}(A') \times_X U, p} \to \kappa(p')$. Note that B' is a local ring. Let $\mathfrak{m} \subset B$ be the maximal ideal. By construction $A \cap \mathfrak{m}$, (resp. $\mathcal{O}_{U,u} \cap \mathfrak{m}$, resp. $A' \cap \mathfrak{m}$) is the maximal ideal of A (resp. $\mathcal{O}_{U,u}$, resp. A'). Set $\mathfrak{q} = B \cap \mathfrak{m}$. This is a prime ideal such that $A \cap \mathfrak{q}$ is the maximal ideal of A. Hence $B_{\mathfrak{q}} \subset \kappa(q')$ is a local ring dominating A. By Algebra, Lemma 00IA we can find a valuation ring $A_1 \subset \kappa(q')$ with field of fractions $\kappa(q')$ dominating $B_{\mathfrak{q}}$. The (local) ring map $\mathcal{O}_{U,u} \to A_1$ gives a morphism $\mathrm{Spec}(A_1) \to U \to X$ such that the diagram

$$\begin{array}{ccccc} \mathrm{Spec}(\kappa(q')) & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(A_1) & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

is commutative. Since the fraction field of A_1 is $\kappa(q')$ and since $\kappa(q')/K$ is finite separable by construction the lemma is proved. □